REAL-TIME SCHEDULING – ANSWER KEY

1(a) Rate Monotonic Scheduling (RMS) (5 Marks)

Rate Monotonic Scheduling is a fixed-priority preemptive scheduling algorithm used in real-time systems for periodic tasks.

- Tasks with shorter period (higher frequency) are assigned higher priority.
- Tasks with longer period are assigned lower priority.
- Priority is fixed and does not change during execution.

Working Principle

- If a higher priority task arrives, it preempts the currently running lower priority task.
- Scheduling decision is based only on task period.

Assumptions of RMS

- All tasks are periodic.
- Deadlines are equal to their periods.
- Tasks are independent (no resource sharing).
- Execution time of tasks is known and constant.
- Context switching time is negligible.

Schedulability Condition

$U = \Sigma (C_i / T_i) \le n(2^{(1/n)} - 1)$
For large n: $U \le 0.693$ (69.3%)

Advantages

- Simple and predictable.
- Easy implementation.
- Suitable for hard real-time systems.

1(b) Earliest Deadline First (EDF) (5 Marks)

EDF is a dynamic priority preemptive scheduling algorithm used in real-time systems.

- The task with the earliest absolute deadline gets highest priority.
- Priorities change dynamically during execution.

Working Principle

- Whenever a new task arrives, scheduler checks deadlines.

- Task with nearest deadline executes first.
- If a new task has earlier deadline, it preempts current task.

Schedulability Condition

$U = \Sigma (C_i / T_i) \leq 1$
CPU utilization can reach 100%.

Advantages

- Higher CPU utilization (up to 100%).
- Optimal for uniprocessor systems.
- Flexible and efficient.
- Works well for both periodic and aperiodic tasks.

1(c) Preemptive vs Non-Preemptive Scheduling (5 Marks)

Feature	Preemptive Scheduling	Non-Preemptive Scheduling
Interruption	Task can be interrupted	Task cannot be interrupted
Response Time	Faster	Slower
Complexity	More complex	Simple
Context Switching	Frequent	Less frequent
Suitability	Hard real-time systems	Soft real-time systems

1(d) Task, Process and Thread (5 Marks)

Entity	Description
Task	Basic schedulable unit in real-time system with execution time, period and deadline.
Process	Independent program in execution with separate memory space.
Thread	Lightweight unit within a process sharing memory with other threads.